Software Development

PRG522

FA1

Due: 30-08-2024

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20231230

Date: 30-08-2024

Question 1

Task 1: Database Design and Normalization

Clubs

Column Name	**Data Type**	**Key**
club_id	INT	Primary Key
club_name	VARCHAR	
stadium	VARCHAR	
coach	VARCHAR	

Players

Column Name	**Data Type**	**Key**
player_id	INT	Primary Key
player_name	VARCHAR	
market_value	DECIMAL	
position	VARCHAR	

Games

Column Name	**Data Type**	**Key**
game_id	INT	Primary Key
date	DATE	
stadium	VARCHAR	

Game Results

Column Name	**Data Type**	**Key**
game_id	INT	Primary Key (PK), Foreign Key (FK) (References Games.game_id)
player_id	INT	Primary Key (PK), Foreign Key (FK) (References Players.player_id)
goals_scored	INT	

Task Two

```
CREATE TABLE Clubs (
   club_id INT PRIMARY KEY,
   club_name VARCHAR(100),
   stadium VARCHAR(100),
   coach VARCHAR(100)
);

CREATE TABLE Players (
   player_id INT PRIMARY KEY,
   player_name VARCHAR(100),
   market_value DECIMAL(15, 2),
   position VARCHAR(50)
);

CREATE TABLE Games (
   game_id INT PRIMARY KEY,
   "date" DATE,
   stadium VARCHAR(100)
);

CREATE TABLE GameResults (
   game_id INT,
   player_id INT,
   goals_scored INT,
   PRIMARY KEY (game_id, player_id),
   FOREIGN KEY (game_id) REFERENCES Games(game_id),
   FOREIGN KEY (player_id) REFERENCES Players(player_id)
);
```

INSERT INTO Clubs (club_id, club_name, stadium, coach) VALUES(1,'FC Barcelona', 'Camp Nou', 'Hansi Flick');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(2,'Real Madrid', 'Santiago Bernabeu', 'Carlo Ancelotti');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(3,'Manchester United', 'Old Trafford', 'Erik ten Hag');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(4, 'Arsenal', 'The Emirates', 'Mikel Arteta');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(5, 'Liverpool','Anfield','Arne Slot');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(6, 'Chelsea', 'Stamford Bridge', 'Enzo Mareca');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(7, 'Napoli', 'Diego Armando Maradona Stadium', 'Antonio Conte');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(8, 'Manchester City', 'Ethihad', 'Pep Guardiola');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(9, 'Bayern Munich', 'Allianz Arena', 'Vincent Kompany');

INSERT INTO Clubs (club_id, club_name, stadium, coach)VALUES(10, 'Paris Saint-Germain','Parc des Princes','Luis Enrique');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(1, 'Lionel Messi', 50000000.00, 'Center Forward');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(2, 'Cristiano Ronaldo', 50000000.00, 'Striker');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(3, 'Bruno Fernandes', 75000000.00, 'Attacking Midfielder');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(4, 'Kevin Debruyne', 85000000.00, 'Attacking Midfielder');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(5, 'Marcus Rashford', 25000000.00, 'Left Midfielder');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(6, 'Jude Bellingham', 50000000.00, 'Attacking Midfielder');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(7, 'Karim Benzema', 20000000.00, 'Center Forward');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(8, 'Mohammed Salah', 95000000.00, 'Right Winger');

```
INSERT INTO Players (player_id, player_name, market_value, position) VALUES(9, 'Robert Lewendowski', 40000000.00, 'Striker');

INSERT INTO Players (player_id, player_name, market_value, position) VALUES(10, 'Erling Haaland', 155000000.00, 'Striker');


INSERT INTO Games VALUES(1, TO_DATE('2022-06-02','YYYY-MM-DD'),'Camp Nou');

INSERT INTO Games VALUES(2, TO_DATE('2004-07-22','YYYY-MM-DD'),'Allianz Arena');

INSERT INTO Games VALUES(3, TO_DATE('2014-12-14','YYYY-MM-DD'),'The Emirates');

INSERT INTO Games VALUES(4, TO_DATE('2024-02-10','YYYY-MM-DD'),'Diego Armando Maradona Stadium');

INSERT INTO Games VALUES(5, TO_DATE('2024-01-20','YYYY-MM-DD'),'Anfield');

INSERT INTO Games VALUES(6, TO_DATE('2020-12-20','YYYY-MM-DD'),'Ethihad');

INSERT INTO Games VALUES(7, TO_DATE('2016-04-10','YYYY-MM-DD'),'Santiago Bernabeu');

INSERT INTO Games VALUES(8, TO_DATE('2012-01-09','YYYY-MM-DD'),'Parc des Princes');

INSERT INTO Games VALUES(9, TO_DATE('2022-01-23','YYYY-MM-DD'),'Old Trafford');

INSERT INTO Games VALUES(10, TO_DATE('2020-03-10','YYYY-MM-DD'),'Stamford Bridge');


INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(6, 10, 3);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(2, 9, 2);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(5, 8, 2);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(4, 7, 4);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(7, 6, 1);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(10, 1, 2);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(9, 2, 3);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(8, 3, 1);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(1, 4, 1);

INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(3, 5, 1);
```

Task 3

```
SELECT
  Players.player_name,
  Players.market_value,
  Clubs.club_name
FROM
  Players
JOIN
  Clubs ON Players.player_id = Clubs.club_id
ORDER BY
  Players.market_value DESC
FETCH FIRST 10 ROWS ONLY;
```

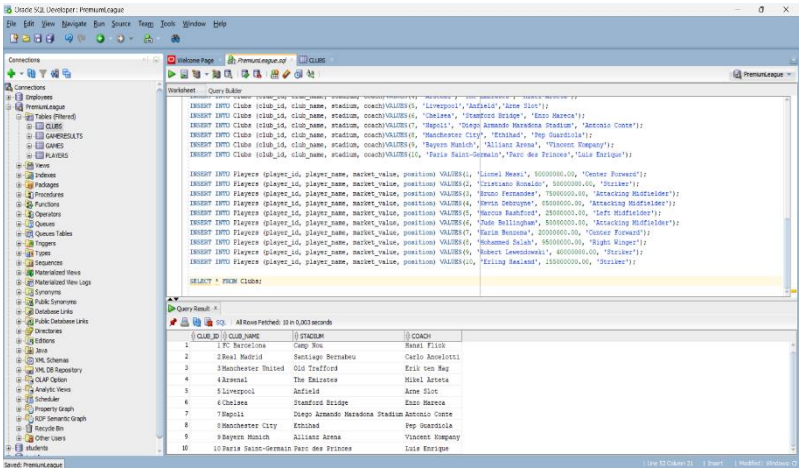
Task 4

i) Database Roles and privileges play a very important role in securing a database, protecting the data inside and restricting access to it. Roles are a preset sets of privileges that can be allocated to users, they help in enforcing the principle of least privilege, where users are granted only the minimum access necessary for their job functions.
Privileges are specific permissions that grant users the ability to perform actions like reading, inserting, updating, and removing data on database objects like tables, views, or procedures. So, in a business Roles and Privileges enforce security policies that restrict access to private data.

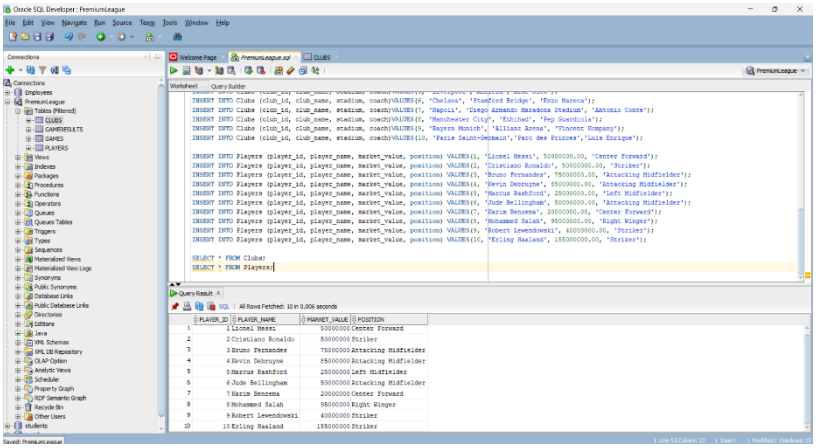
ii) Grant Options allow the Database Administrator to control access to a, being able to assign privileges to a user. Available Grant options are granting privileges to users like granting a user permission to access a database, and taking away that grant when they are done working/accessing that database, this is done by the DBA.

iii) In a database views are like virtual tables which present data to users, this is usually data that the user themself wanted to see and wrote a query to see it and it appears as a view. So, we can say that views can hide complex details of a databases' tables from users who do not need to see them, since with a few you can just show them what they need to only and nothing more.

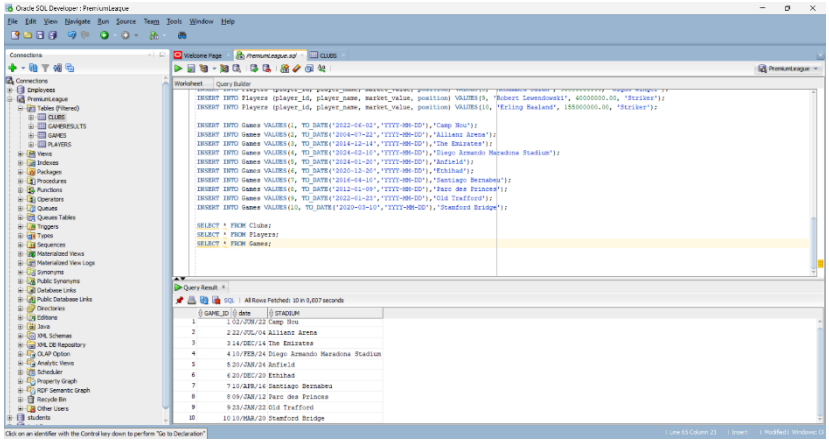
Screenshots



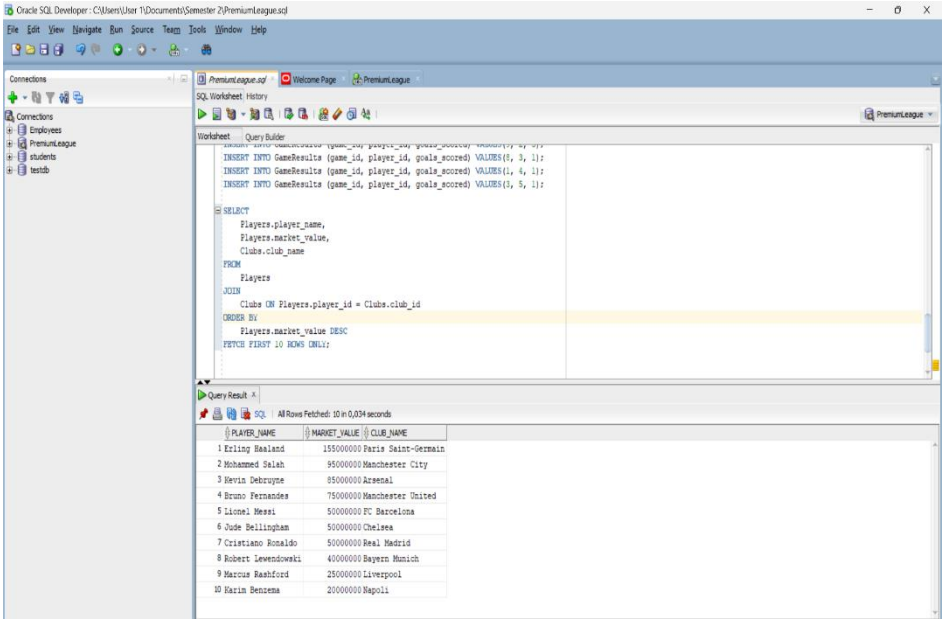
Clubs Table created and date inserted



Player table created and data insrted



Games Tavle created and data inserted


Oracle SQL Developer : C:\Users\User 1\Documents\Semester 2\PremiumLeague.sql
File Edit View Navigate Run Source Team Tools Window Help
Connections
Connections
Employees
PremiumLeague
students
testdb
PremiumLeague.sql
Welcome Page
PremiumLeague
SQL Worksheet History
Worksheet
Query Builder
INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(5, 3, 1);
INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(1, 4, 1);
INSERT INTO GameResults (game_id, player_id, goals_scored) VALUES(3, 5, 1);
SELECT
Players.player_name,
Players.market_value,
Clubs.club_name
FROM
Players
JOIN
Clubs ON Players.player_id = Clubs.club_id
ORDER BY
Players.market_value DESC
FETCH FIRST 10 ROWS ONLY;
Query Result
All Rows Fetched: 10 in 0,034 seconds
PLAYER_NAME MARKET_VALUE CLUB_NAME
1 Erling Haaland 155000000 Paris Saint-Germain
2 Mohammed Salah 95000000 Manchester City
3 Kevin Debruyne 85000000 Arsenal
4 Bruno Fernandes 75000000 Manchester United
5 Lionel Messi 50000000 FC Barcelona
6 Jude Bellingham 50000000 Chelsea
7 Cristiano Ronaldo 50000000 Real Madrid
8 Robert Lewendowski 40000000 Bayern Munich
9 Marcus Rashford 25000000 Liverpool
10 Karim Benzema 20000000 Napoli

Task 3 Query